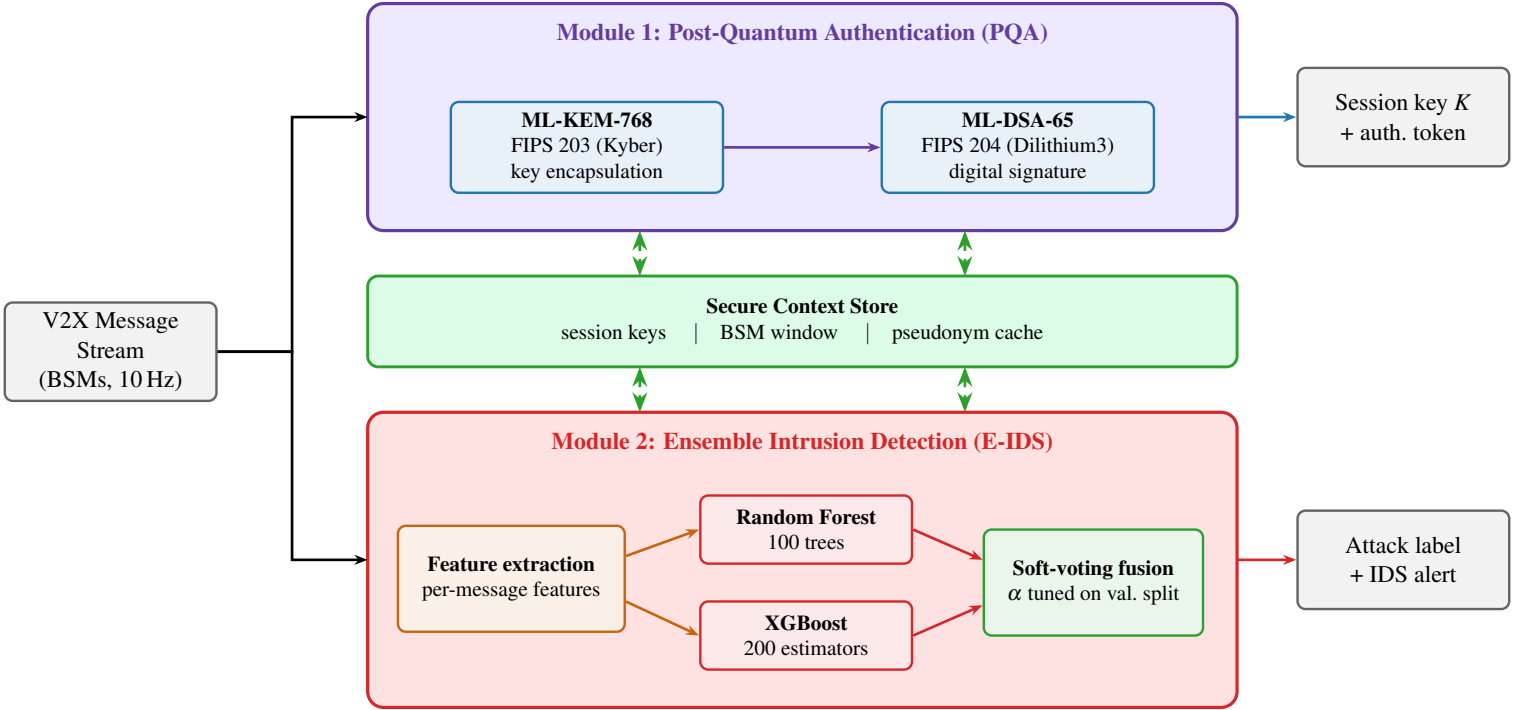
Module 1: Post-Quantum Authentication (PQA)
ML-KEM-768
FIPS 203 (Kyber)
key encapsulation
ML-DSA-65
FIPS 204 (Dilithium3)
digital signature
Session key K
+ auth. token
V2X Message
Stream
(BSMs, 10 Hz)
Secure Context Store
session keys | BSM window | pseudonym cache
Module 2: Ensemble Intrusion Detection (E-IDS)
Feature extraction
per-message features
Random Forest
100 trees
XGBoost
200 estimators
Soft-voting fusion
α tuned on val. split
Attack label
+ IDS alert